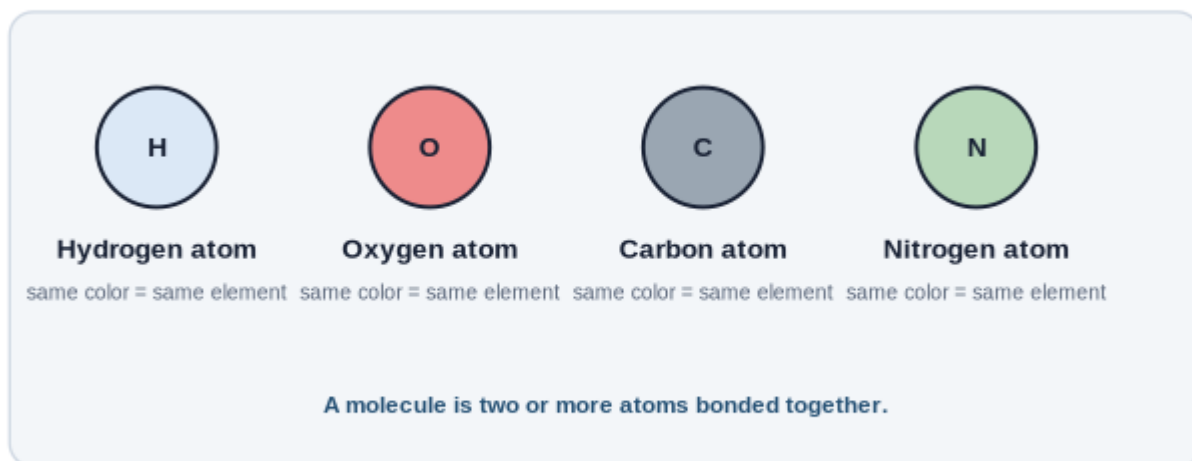


Intro Molecules - Student Handout

7th Grade Science Learning target: I can make and draw a model of molecules showing the composition before and after a chemical change as well as what happens when temperature changes.

1. Model key

Model key: circles stand for atoms

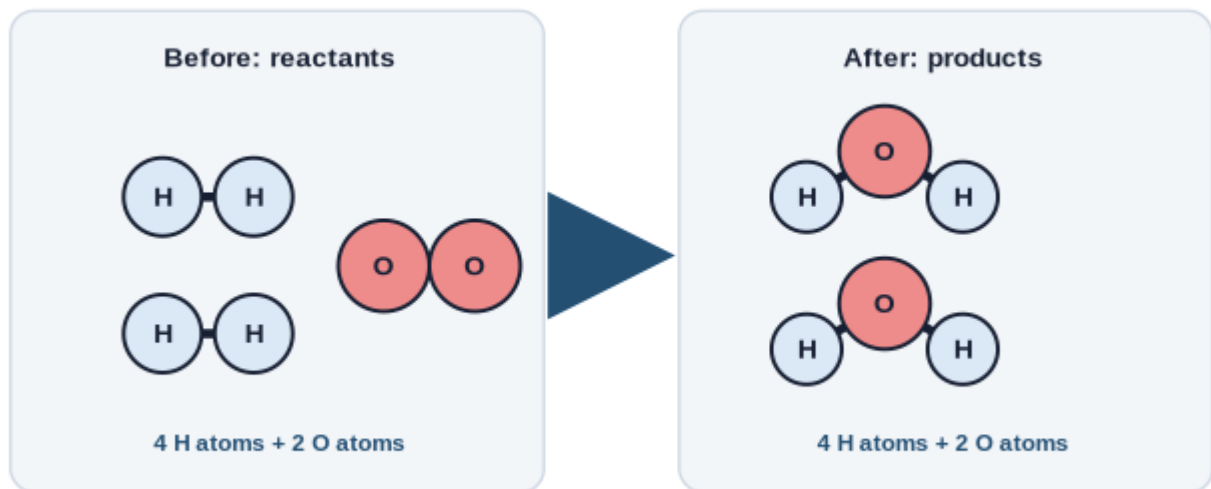


Use the same atom key before and after a change.

Remember: same color/letter = same element. Connected circles = atoms bonded in a molecule.

2. Chemical change model

Chemical change model: atoms rearrange



In a chemical change, atoms rearrange into new molecules. The atom counts should still match.

Question

Your notes

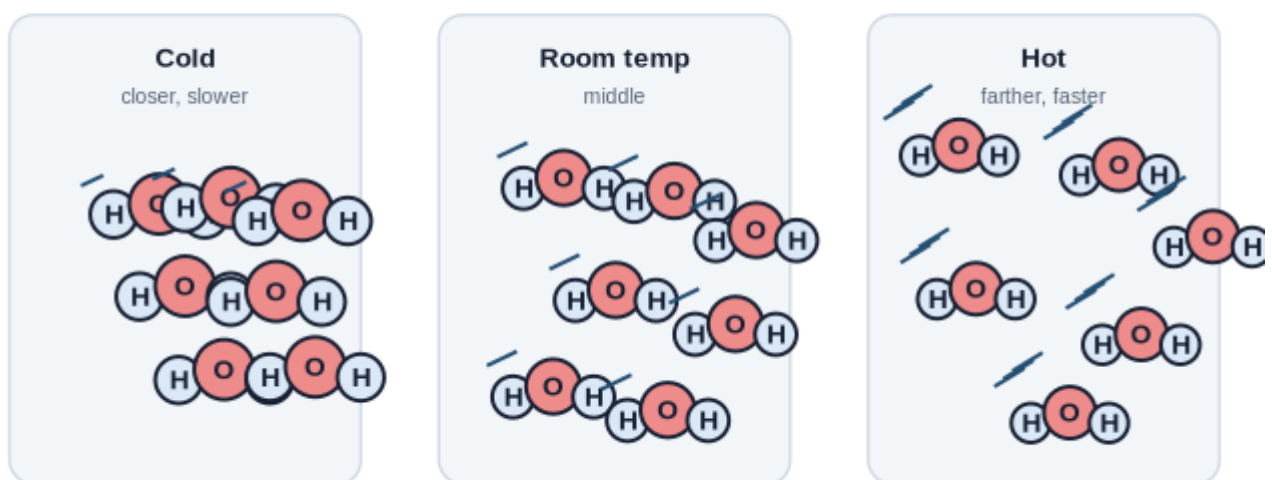
What atom types are present before?

What atom types are present after?

What changed in the molecule arrangement?

3. Temperature change model

Temperature change model: same molecules, different motion



Heating does not turn water molecules into different molecules.

Heating changes molecular motion and spacing. It does not automatically make new molecules.

Heating: molecules move faster and tend to be farther apart.

Cooling: molecules move slower and tend to be closer together.

4. Practice drawing

A. Draw a before-and-after model for: 2 H₂ molecules + 1 O₂ molecule → 2 H₂O molecules.

H atoms O atoms

Before

After

B. Explain: How is a chemical-change model different from a temperature-change model?